SINKGRAB · PROJECT PLAN

SinkGrab project plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SKG-PLN-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/11-plan.md @ 780e4b8		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (O2 to O5) recorded; O1 still open; CalRig tasks for the slip clutch and load indicator; cost USD 3,063
0.1	2026-10-24	Amish Chadha	First project plan with the reference list (kit 1.9.1 upgrade, stage P3)

SinkGrab project plan

SinkGrab is at TRL 3 and stage P3 (design outputs): the detailed model, drawings, priced BOM, build plan and decisions register exist for option A, and option B is modelled to the same detail. The next gate is P4 (verification, TRL 4), on hold under the portfolio cap; before it Amish is asked to decide the concept (SKG-DEC-001 O1), how the rings are helped to sink (O2), the two new requirements (O3), the overload device and proof basis (O4) and the scraper turning force (O5). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and requirements	Amish (owner)	Concept choice (O1); PR-012 and PR-013 accepted 2026-10-25 (O3)	Concept choice awaiting Amish
Design and model	Design lead	Detailed D3 model of both options; drawings, STEP and STL of option A	Done on paper (2026-10-24)
Calculations and STEM check	Design lead	SKG-CAL-001 v0.6; SKG-STM-001 v0.3 with two rows failing (option B, with O1) and thirteen open until trials, after Amish's decisions of 2026-10-25	Done on paper; open rows need TRL 4
Build plan	Design lead	SKG-BLD-001 for option A with the kit 1.9.1 fixes	Done on paper
Co-design partner	Amish (owner)	Agreement with the Kerala Ground Water Department or the Ethiopian NGO	Not started
Proof loading	Test lead (CalRig)	Proof-load record for the line, cradle, capstan and tripod	On hold (TRL 4)
Trials	Co-design partner with the design lead	Bite trial, timed cycle trial, fold check, first well	On hold (TRL 4 and 5)
Patent screen	Amish (owner)	Targeted search for under-curb scraper claims	Before public release

Resources needed

Table 2. Resources (P4 and later on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
Partner workshop	Build of the option A kit from the drawings	About 2 to 3 weeks for two people (estimate)	3,063 in parts (BOM); labour to assess
CalRig and a load cell	Proof load of 188 kg (1.25 x the rated load); slip clutch setting on each rope layer; load indicator calibration	2 days (estimate)	To assess
Test pit of saturated sand, laterite and clayey sand	Bite and fill per soil (PR-003, PR-013); option B fill if chosen	1 week (estimate)	To assess
Partner well and crew	Timed cycle (PR-004), sinking and skin friction (PR-001), tilt (PR-006)	2 weeks (estimate)	To assess
Ergonomics reviewer	ISO 11228-2 assessment of the T-bar (STEM B4)	2 days (estimate)	To assess
Regulatory reviewer in the trial country	Lifting gear, gas testing and well permission rules (SKG-REG-001)	2 days (estimate)	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (2026-10-24 for the pipeline documents)
P1 Definition	URS, PRS, RSK, MKT	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR	Met; the concept selection is awaiting Amish (SKG-DDR-005)
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Met on paper (2026-10-24)
P4 Verification	Proof-load and lab test reports, verification report	On hold
P5 Validation	Partner well trials, human factors	On hold
P6 Commercialization	Open route with a named deployment partner	On hold

Next steps and decision requested

1. Amish decides O1 (concept: A, the current design, or B). O2 to O5 were decided on 2026-10-25 (bentonite annulus, PR-012 and PR-013, slip clutch and load indicator with proof at 1.25 x the rated load, four people at the T-bar).
2. The coordinator renders `media/options.png` and the product renders (hero, studio) in Cycles from `cad/src/model.py` and `cad/src/product_model.py`.

3. When the portfolio phase allows TRL 4: build, set the slip clutch and calibrate the load indicator on CalRig, proof-load, fold the scraper through a 0.8 m ring, then the bite and timed cycle trials and the bentonite annulus in the test pit before a partner well.

References

- Abbott. *Hand Dug Wells: Choice of Technology and Construction Manual*. https://sswm.info/sites/default/files/reference_attachments/ABBOT%204000%20Hand%20Dug%20Well%20Manual.pdf.
- Akvopedia. "Traditional Hand-Dug Wells." https://akvopedia.org/wiki/Traditional_hand-dug_wells.
- American Water Works Association. *AWWA C654-13, Disinfection of Wells*.
- British Geological Survey. "Study Shows Boreholes Are Key to Drought Resilience in Ethiopia." https://www.bgs.ac.uk/news/study-shows-boreholes-are-key-to-drought-resilience-in-ethiopia/.
- Bureau of Indian Standards. *IS 10500:2012, Drinking Water, Specification*; *IS 3955:1967, Code of Practice for Design and Construction of Well Foundations*.
- Central Ground Water Authority. *Guidelines to Regulate and Control Ground Water Extraction in India*, S.O. 3289(E), September 24, 2020.
- Construction Civil. "Well Sinking: Tilt and Shift in Well Foundation." https://www.constructioncivil.com/well-sinking-tilt-shift-in-well-foundation/.
- CO2Meter. "Portable 5% CO2 Safety Detector." https://co2meter.com/products/personal-co2-monitor (checked 2026-10-25).
- Deccan Chronicle. "Thiruvananthapuram: Well Diggers Strike Gold in Summer." 2017. https://www.deccanchronicle.com/nation/in-other-news/040517/thiruvananthapuram-well-diggers-strike-gold-in-summer.html.
- Henderson, Bob. "Measuring Carbon Dioxide Levels." *Safety+Health*, September 1, 2010. https://www.safetyandhealthmagazine.com/articles/measuring-carbon-dioxide-levels-2.
- Inter Aide and Réseau Pratiques. *Rehabilitation of Hand-Dug Wells: Diagnostic and Technical Solutions*. 2012.
- Messer, Caleb. "Rope and Rigging Compatibility, Part Two of Two." *Inside Towers*, September 6, 2019.
- Network Rail. "Lifting and Handling in Teams." https://safety.networkrail.co.uk/safety/manual-handling/principles-of-manual-handling/lifting-and-handling-in-teams/.
- Our World in Data. "What Does It Mean to Not Have Access to Safe Water?" https://ourworldindata.org/what-no-safe-water-means.
- Oxfam WASH. "Repairing, Cleaning and Disinfecting Hand Dug Wells." https://www.oxfamwash.org/repairing-cleaning-and-disinfecting-hand-dug-wells/.
- Sal Marine. "Honda WB20 Water Pump, WB20XT4DRX." https://shop.salmarine.com/honda-wb20-water-pump-wb20xt4drx.html (checked 2026-10-25).
- SOPPECOM. *Groundwater Resource and Governance in Kerala*. https://www.soppecom.org/pdf/Groundwater-Resource-and-Governance-in-Kerala.pdf.
- Terzaghi, Karl, Ralph B. Peck and Gholamreza Mesri. *Soil Mechanics in Engineering Practice*. 3rd ed. New York: Wiley, 1996.
- The Tribune. "4 Die of Asphyxiation after Inhaling Toxic Gas inside Well in Bihar's Gaya Ji." 2026. https://www.tribuneindia.com/news/india/4-die-of-asphyxiation-after-inhaling-toxic-gas-inside-well-in-bihars-gaya-ji/.
- WaterAid. *Technology Brief: Hand-Dug Wells*. 2013. https://sswm.info/sites/default/files/reference_attachments/WATERAID%202013.%20Hand-dug%20wells.pdf.

- World Health Organization. *Guidelines for Drinking-water Quality*. 4th ed. incorporating the first and second addenda. Geneva: WHO, 2022.
- ZimLive. "3 Killed in Carbon Monoxide Poisoning after Turning on Generator in a Well." 2023. https://www.zimlive.com/3-killed-in-carbon-monoxide-poisoning-after-turning-on-generator-in-a-well/.
- Patents: CN101787703B and US1750728A, see SKG-PRB-001.
- Standards cited by number in SKG-REG-001: EN 13157, EN 13889, EN ISO 10547, ISO 1141, ISO 2307, ISO 11228-1 and -2, EN 60079-29-1, EN 45544, OSHA 29 CFR 1910.146, LOLER 1998, Directive 2006/42/EC, Regulation (EU) 2023/1230, EN 10219, EN 10025-2, ISO 4017, ISO 4032, ISO 7089, ISO 2341, ISO 1234.